

Stationary distributions

April 23, 2026

Contents

Stationary distributions

Balance equations

Examples

- A random walk with barriers

- Random walks on graphs

Mean first passage and mean recurrence times

Stationary distributions

Definition

A vector (if the state space S is finite) or a sequence (if S is countable) of real numbers

$$(\pi_j : j \in S)$$

is a stationary distribution of the Markov chain if:

- ▶ $\pi_j \geq 0$ for all j and $\sum_{j \in S} \pi_j = 1$.
- ▶ The following equality holds:

$$\pi_j = \sum_{i \in S} \pi_i p_{ij}, \quad j \in S$$

Stationary distributions

If the initial distribution ($\mathbb{P}(X_0 = j): j \in S$) is a stationary distribution ($\pi_j: j \in S$), then

$$\begin{aligned}\mathbb{P}(X_1 = j) &= \sum_{i \in S} \mathbb{P}(X_1 = j \mid X_0 = i) \mathbb{P}(X_0 = i) \\ &= \sum_{i \in S} p_{ij} \pi_i = \pi_j\end{aligned}$$

Therefore, for all $n \geq 0$,

$$\mathbb{P}(X_n = j) = \pi_j, \quad j \in S$$

- ▶ All random variables X_n share an identical probability distribution, indicating that the probability of the Markov chain occupying a particular state remains constant over time.

Stationary distributions

If the Markov chain is finite with $|S| = r$ states, a stationary distribution $\pi = (\pi_1, \dots, \pi_r)$ corresponds to a left eigenvector of the transition matrix $\mathbf{P}$, with an eigenvalue equal to 1:

$$\pi = \pi \mathbf{P}$$

This equality implies that, for all $n \geq 0$,

$$\pi = \pi \mathbf{P}^n$$

Balance equations

Let $(\pi_j: j \in S)$ be a stationary distribution. The balance equations $\pi_j = \sum_{i \in S} \pi_i p_{ij}$ can be reformulated as

$$\pi_j = \pi_j p_{jj} + \sum_{i \neq j} \pi_i p_{ij}, \quad j \in S$$

and, since $1 - p_{jj} = \sum_{k \neq j} p_{jk}$:

(Balance equations)

For all $j \in S$,

$$\pi_j \sum_{k \neq j} p_{jk} = \sum_{i \neq j} \pi_i p_{ij}$$

Balance equations

(Balance equations)

- ▶ *The left-hand side:*

$$\pi_j \sum_{k \neq j} p_{jk}$$

represents the probability flow out of state j .

- ▶ *In contrast, the right-hand side:*

$$\sum_{i \neq j} \pi_i p_{ij}$$

corresponds to the total probability flow into state j .

- ▶ *A stationary distribution establishes equilibrium between these two probability flows.*

Balance equations

Example:

Consider a finite chain with state space $S = \{1, 2, 3\}$ and transition probability matrix

$$P = \begin{pmatrix} 0.2 & 0.7 & 0.1 \\ 0.1 & 0.8 & 0.1 \\ 0.4 & 0.4 & 0.2 \end{pmatrix}$$

Let us prove that the chain admits a stationary distribution.

Balance equations

The equation $\pi = \pi P$ corresponds to the system

$$\pi_1 = 0.2 \pi_1 + 0.1 \pi_2 + 0.4 \pi_3$$

$$\pi_2 = 0.7 \pi_1 + 0.8 \pi_2 + 0.4 \pi_3$$

$$\pi_3 = 0.1 \pi_1 + 0.1 \pi_2 + 0.2 \pi_3$$

This system can be reformulated as (balance equations):

$$0.8 \pi_1 = 0.1 \pi_2 + 0.4 \pi_3$$

$$0.2 \pi_2 = 0.7 \pi_1 + 0.4 \pi_3$$

$$0.8 \pi_3 = 0.1 \pi_1 + 0.1 \pi_2$$

Balance equations

The unique solution of the balance equations that satisfies the normalization condition

$$\pi_1 + \pi_2 + \pi_3 = 1$$

is the probability vector

$$\pi = \left(\frac{4}{27}, \frac{20}{27}, \frac{3}{27} \right) \approx (0.148, 0.741, 0.111)$$

which is the stationary distribution of the Markov chain.

Balance equations

Example:

Sometimes, we can benefit from the symmetry of the chain.

For instance, if

$$\mathbf{P} = \begin{pmatrix} 1/2 & 1/4 & 1/4 \\ 1/2 & 0 & 1/2 \\ 1/4 & 1/4 & 1/2 \end{pmatrix}$$

we can assume

$$\pi_1 = \pi_3$$

Balance equations

For example, if $(1, t, 1)$ is a vector that satisfies the balance equations, we must have:

$$(1 \ t \ 1) = (1 \ t \ 1) \mathbf{P}$$

So, we deduce that $t = 1/2$. We conclude that the vector

$$\left(1, \frac{1}{2}, 1\right)$$

is a solution.

By normalizing this vector so that the sum of its components equals 1, we obtain the stationary distribution:

$$\pi = \left(\frac{2}{5}, \frac{1}{5}, \frac{2}{5}\right)$$

Existence of a stationary distribution

Theorem

An irreducible Markov chain has a stationary distribution if and only if it is non-null recurrent (i.e., all the states are positive).

In that case:

- ▶ *The stationary distribution $(\pi_j: j \in S)$ is unique.*
- ▶ *For each $j \in S$,*

$$\pi_j = \frac{1}{\mu_j} > 0,$$

where μ_j is the mean recurrence time of the state j .

Corollary

A finite irreducible Markov chain has a unique stationary distribution.

Long-term expected fraction of time

Let $v_{ij}(n)$ be the expected number of visits to state j within the first n transitions, starting from state i .

Theorem

Let $(\pi_j = 1/\mu_j: j \in S)$ be the stationary distribution of an irreducible and non-null recurrent chain. Therefore, for all $j \in S$,

$$\pi_j = \lim_{n \rightarrow \infty} \frac{v_{ij}(n)}{n}, \quad \text{independently of } i.$$

- Thus, $\pi_j = 1/\mu_j$ is the long-term expected fraction of time that the state is equal to j .

Detailed balance equations

Theorem

Let $(\pi_j \geq 0 : j \in S)$ be such that $\sum_{j \in S} \pi_j = 1$.

If for all $i, j \in S$

$$\pi_i p_{ij} = \pi_j p_{ji}$$

then $(\pi_j : j \in S)$ is a stationary distribution.

Proof: We must show that if $\pi_i p_{ij} = \pi_j p_{ji}$, then the balance equations are satisfied:

$$\sum_{i \in S} \pi_i p_{ij} = \sum_{i \in S} \pi_j p_{ji} = \pi_j \sum_{i \in S} p_{ji} = \pi_j$$

Doubly stochastic matrices

An stochastic matrix $\mathbf{P} = (p_{ij})$ is **doubly stochastic** if $\sum_i p_{ij} = 1$.
(The sum of the elements of each column is also 1.)

Theorem

If the transition probability matrix of a finite and irreducible Markov chain (with r states) is doubly stochastic, then the stationary distribution is uniform:

$$\pi_j = \frac{1}{r}, \quad 1 \leq j \leq r$$

Doubly stochastic matrices

Proof:

Firstly, notice that

$$\sum_{j=1}^r \pi_j = \sum_{j=1}^r \frac{1}{r} = 1$$

Moreover, the balance equations are satisfied:

$$\sum_{i=1}^r \pi_i p_{ij} = \sum_{i=1}^r \frac{1}{r} p_{ij} = \frac{1}{r} \sum_{i=1}^r p_{ij} = \frac{1}{r} = \pi_j$$

Doubly stochastic matrices

Example:

Consider a Markov chain such that $S = \{1, 2, 3, 4, 5\}$ and

$$\mathbf{P} = \begin{pmatrix} 0 & p & 0 & 0 & 1-p \\ 1-p & 0 & p & 0 & 0 \\ 0 & 1-p & 0 & p & 0 \\ 0 & 0 & 1-p & 0 & p \\ p & 0 & 0 & 1-p & 0 \end{pmatrix}$$

The chain is finite and irreducible. Moreover, $\mathbf{P}$ is doubly stochastic. Hence,

$$\pi = \left(\frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5} \right)$$

A random walk with barriers

Consider a Markov chain such that $S = \{0, 1, \dots, m\}$ and

$$p_{ij} = \begin{cases} 1 - b, & j = i = 0 \\ b, & j = i + 1, 0 \leq i \leq m - 1 \\ d, & j = i - 1, 1 \leq i \leq m \\ 1 - b - d, & 1 \leq j = i \leq m - 1 \\ 1 - d, & j = i = m \end{cases}$$

(We assume that $b > 0$, $d > 0$ and $b + d \leq 1$.)

- The chain is finite and irreducible. Therefore, there is a unique stationary distribution.

A random walk with barriers

In this example, the detailed balance equations are fulfilled (Why?).

Hence,

$$\pi_i b = \pi_{i+1} d, \quad 0 \leq i \leq m-1$$

Let $\rho = b/d$. Then

$$\pi_{i+1} = \pi_i \rho, \quad 0 \leq i \leq m-1$$

Therefore,

$$\pi_i = \pi_0 \rho^i, \quad 0 \leq i \leq m$$

A random walk with barriers

The value π_0 is determined by the condition $\sum_{i=0}^m \pi_i = 1$. We have:

$$1 = \pi_0(1 + \rho + \cdots + \rho^m) = \begin{cases} \pi_0 \frac{1 - \rho^{m+1}}{1 - \rho}, & \text{if } \rho \neq 1 \\ \pi_0 (m + 1), & \text{if } \rho = 1 \end{cases}.$$

Consequently, for $0 \leq i \leq m$:

$$\pi_i = \begin{cases} \frac{(1 - \rho) \rho^i}{1 - \rho^{m+1}}, & \text{if } \rho \neq 1 \\ \frac{1}{m + 1}, & \text{if } \rho = 1 \end{cases}$$

A random walk with barriers

Some comments:

- ▶ If $\rho = 1$ ($b = d$), the transition probability matrix is doubly stochastic and we have the uniform distribution.
- ▶ If $\rho < 1$ ($b < d$), then

$$\lim_{m \rightarrow \infty} \pi_i = (1 - \rho) \rho^i$$

for any fixed $i \geq 0$.

- ▶ On the contrary, if $\rho > 1$ ($b > d$), then

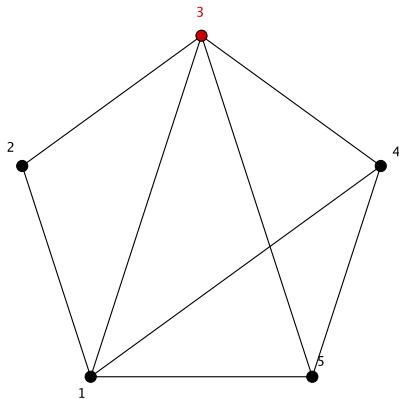
$$\lim_{m \rightarrow \infty} \pi_i = 0$$

Random walks on graphs

Let $G = (V, E)$ be a connected graph with $|V| = n$ vertices and $|E| = m$ edges. For simplicity, let $V = \{1, 2, \dots, n\}$.

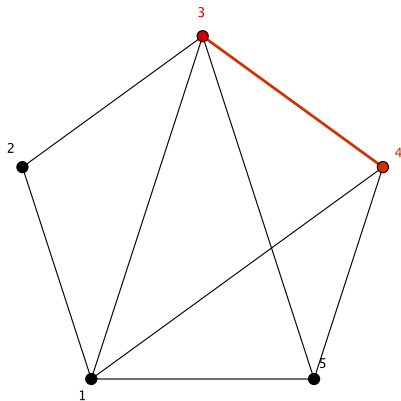
- ▶ Given an initial vertex v_0 , select **at random** an adjacent vertex v_1 , and move from v_0 to v_1 .
- ▶ In the next step, select **at random** a neighbour v_2 of v_1 , and move to it.
- ▶ By repeating this procedure, the sequence of visited vertices, $v_0, v_1, v_2, \dots, v_k, \dots$, is called a **random walk** on G .

Random walks on graphs



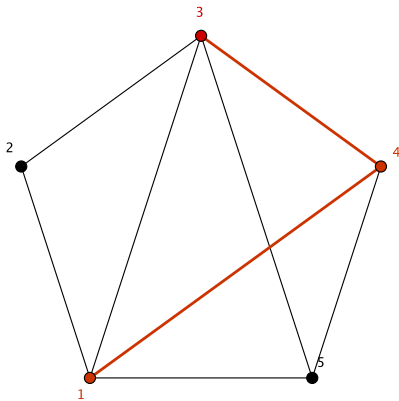
$$v_0 = 3$$

Random walks on graphs



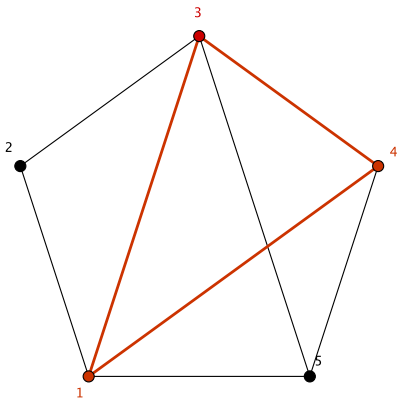
$3 \rightarrow 4$

Random walks on graphs



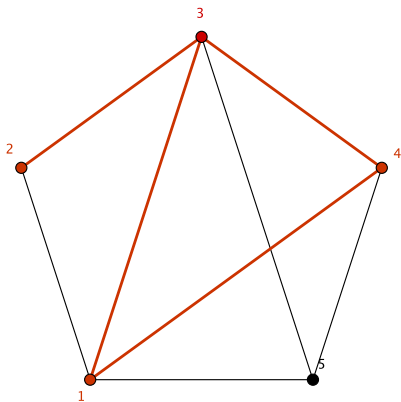
$3 \rightarrow 4 \rightarrow 1$

Random walks on graphs



$3 \rightarrow 4 \rightarrow 1 \rightarrow 3$

Random walk on a graph



$3 \rightarrow 4 \rightarrow 1 \rightarrow 3 \rightarrow 2 \rightarrow \dots$

Random walks on graphs

- ▶ At each step k , we have a random variable X_k that takes values on the set V . Hence, the random sequence $\{X_k: k \geq 0\}$ is a discrete-time stochastic process with state space V .
- ▶ What does **at random** mean? If $X_k = i$, **uniformly** choose an adjacent vertex j to move to.

Let d_i denote the number of edges incident to vertex i . Then,

$$p_{ij} = \mathbb{P}(X_{k+1} = j \mid X_k = i) = \begin{cases} \frac{1}{d_i}, & \text{if } ij \in E \\ 0, & \text{otherwise} \end{cases}$$

Random walks on graphs

Notice that the **Markov property** holds:

$$\begin{aligned}\mathbb{P}(X_{k+1} = j \mid X_k = i, X_{k-1} = i_{k-1}, \dots, X_0 = i_0) \\ = \mathbb{P}(X_{k+1} = j \mid X_k = i) \equiv p_{ij}\end{aligned}$$

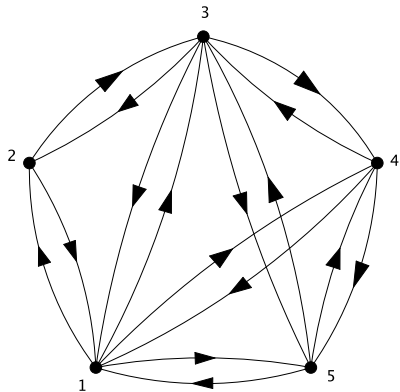
The sequence of vertices visited by the random walk,

$$X_0, X_1, \dots, X_n, \dots$$

constitutes a finite Markov chain with state space V and transition probability matrix:

$$\mathbf{P} = (p_{ij}), \quad i, j \in V$$

Random walks on graphs



$$P = \begin{pmatrix} 0 & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 \\ \frac{1}{4} & \frac{1}{4} & 0 & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{3} & 0 & \frac{1}{3} & 0 & \frac{1}{3} \\ \frac{1}{3} & 0 & \frac{1}{3} & \frac{1}{3} & 0 \end{pmatrix}$$

Random walks on graphs

It is easy to find a probability vector satisfying the detailed balance equations:

$$\frac{\pi_i}{d_i} = \frac{\pi_j}{d_j}, \quad ij \in E$$

Indeed, because G is connected, we conclude that

$$\frac{\pi_i}{d_i} = k, \quad \text{for all } i \in V$$

Furthermore, since

$$1 = \sum_{i \in V} \pi_i = k \sum_{i \in V} d_i = k 2m$$

we must have $k = 1/(2m)$.

Random walks on graphs

(Stationary distribution)

- ▶ *The stationary probabilities are proportional to the degrees of the vertices. More precisely,*

$$\pi_i = \frac{d_i}{2m}, \quad \text{for all } i \in V$$

- ▶ *In particular, if G is d -regular (all the vertices have the same degree d), then*

$$\pi_i = \frac{d}{2m} = \frac{1}{n}, \quad \text{for all } i \in V$$

In this case, π is the uniform distribution.

Mean first passage times

Let m_{ij} represent the expected number of steps until the chain first visits state j , given that it starts in state i . Hence, if $j \neq i$ and $T_j = \min\{n \geq 1: X_n = j\}$, then

$$m_{ij} = \mathbb{E}(T_j | X_0 = i)$$

(We take $m_{ii} = 0$ if $j = i$.)

Let $j \neq i$. By conditioning on the first step, we have:

$$\begin{aligned}\mathbb{E}(T_j | X_0 = i) &= \sum_{k \in S} \mathbb{E}(T_j | X_1 = k, X_0 = i) \mathbb{P}(X_1 = k | X_0 = i) \\ &= \sum_{k \in S} \mathbb{E}(T_j | X_1 = k) \mathbb{P}(X_1 = k | X_0 = i)\end{aligned}$$

Mean first passage times

Therefore,

For all $i, j \in S$, $i \neq j$, we have $m_{ij} = p_{ij} + \sum_{k \neq j} p_{ik}(m_{kj} + 1)$.

Equivalently, since $\sum_{k \in S} p_{ik} = 1$, $m_{ij} = 1 + \sum_{k \neq j} p_{ik}m_{kj}$.

Analogously, for the mean recurrence time, $\mu_i = \mathbb{E}(T_i | X_0 = i)$:

For all $i \in S$, $\mu_i = \sum_{k \in S} p_{ik}(m_{ki} + 1) = 1 + \sum_{k \in S} p_{ik}m_{ki}$.

Mean first passage and recurrence times (matrix analysis)

Let us consider a finite and irreducible Markov chain with r states.

Let

- ▶ $\mathbf{M}$ be a square $r \times r$ matrix such that $(\mathbf{M})_{ij} = m_{ij}$ (with $m_{ii} = 0$).
- ▶ $\mathbf{D} = \text{diag}(\mu_1, \dots, \mu_r)$.
- ▶ $\mathbf{C}$ be a square $r \times r$ matrix with all its entries equal to 1.

The equations relating mean first passage times and mean recurrence times can be written as $\mathbf{M} + \mathbf{D} = \mathbf{C} + \mathbf{P}\mathbf{M}$.

That is,

$$(\mathbf{I} - \mathbf{P})\mathbf{M} = \mathbf{C} - \mathbf{D}$$

Mean first passage and recurrence times (matrix analysis)

Let π be the stationary distribution vector.

Notice that $\pi = \pi P$ implies $\pi(I - P) = \mathbf{0}$. Hence,

$$\mathbf{0} = \pi(I - P)M = \pi(C - D) = \pi C - \pi D$$

Therefore, since

$$\pi C = (1, \dots, 1), \quad \pi D = (\pi_1 \mu_1, \dots, \pi_r \mu_r)$$

we deduce that

$$\pi_i = \frac{1}{\mu_i} > 0, \quad 1 \leq i \leq r$$

Mean first passage and recurrence times (matrix analysis)

Finally, let $\mathbf{\Pi}$ be a square matrix with all its rows equal to π .

Theorem

For a finite and irreducible Markov chain, the following results hold:

- (a) *The matrix $\mathbf{I} - \mathbf{P} + \mathbf{\Pi}$ has an inverse $\mathbf{Z}$ known as the fundamental matrix.*
- (b) *The mean first passages times m_{ij} are derived from $\mathbf{Z} = (z_{ij})$ and the stationary distribution vector π by the expressions:*

$$m_{ij} = \frac{z_{jj} - z_{ij}}{\pi_j}$$